

Description and choice data for the domain “Coffee”

Description of the choice domain 31, Coffee

The prompt question and the universe of five response options in the choice domain **Coffee** are as follows. The labels a , b , c , d and e were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

The source for this domain is the website buycoffeecanada.com, with prices in Canadian dollars.

You need to buy 16oz of ground coffee for a brunch with friends. Which one of the following ground coffees would you choose?

Price	Fair Trade	Name: Description
18.71	Yes	Ethiopian Yirgacheffe: vibrant and intensely aromatic, fruity
9.99	No	Colombian Supremo: mellow cup, complex aromas and rich flavours
13.72	Yes	Colombian Organic: medium body, fragrant aroma and mild acidity
12.35	No	Tanzania Peaberry: full bodied coffee, chocolatey aroma, wine-like finish
13.46	No	Sumatra Mandheling: exotic, earthy, bright with low acidity

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

Coffee

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Figure 1: Screenshot for domain Coffee

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	8	32	-	-	-	0.200	0.800	-	-	-
$\{a, c\}$	7	-	33	-	-	0.175	-	0.825	-	-
$\{b, c\}$	-	23	17	-	-	-	0.575	0.425	-	-
$\{a, b, c\}$	1	14	25	-	-	0.025	0.350	0.625	-	-
$\{a, d\}$	16	-	-	24	-	0.400	-	-	0.600	-
$\{b, d\}$	-	30	-	10	-	-	0.750	-	0.250	-
$\{a, b, d\}$	5	25	-	10	-	0.125	0.625	-	0.250	-
$\{c, d\}$	-	-	27	14	-	-	-	0.659	0.341	-
$\{a, c, d\}$	6	-	26	8	-	0.150	-	0.650	0.200	-
$\{b, c, d\}$	-	23	12	5	-	-	0.575	0.300	0.125	-
$\{a, b, c, d\}$	2	18	15	5	-	0.050	0.450	0.375	0.125	-
$\{a, e\}$	19	-	-	-	21	0.475	-	-	-	0.525
$\{b, e\}$	-	34	-	-	6	-	0.850	-	-	0.150
$\{a, b, e\}$	7	28	-	-	5	0.175	0.700	-	-	0.125
$\{c, e\}$	-	-	33	-	7	-	-	0.825	-	0.175
$\{a, c, e\}$	9	-	20	-	11	0.225	-	0.500	-	0.275
$\{b, c, e\}$	-	17	19	-	4	-	0.425	0.475	-	0.100
$\{a, b, c, e\}$	3	17	18	-	2	0.075	0.425	0.450	-	0.050
$\{d, e\}$	-	-	-	19	21	-	-	-	0.475	0.525
$\{a, d, e\}$	8	-	-	18	14	0.200	-	-	0.450	0.350
$\{b, d, e\}$	-	32	-	7	2	-	0.780	-	0.171	0.049
$\{a, b, d, e\}$	8	23	-	6	3	0.200	0.575	-	0.150	0.075
$\{c, d, e\}$	-	-	25	7	8	-	-	0.625	0.175	0.200
$\{a, c, d, e\}$	4	-	21	9	6	0.100	-	0.525	0.225	0.150
$\{b, c, d, e\}$	-	20	14	6	0	-	0.500	0.350	0.150	0.000
$\{a, b, c, d, e\}$	4	17	14	2	3	0.100	0.425	0.350	0.050	0.075

Table 1: Observed choice counts and proportions.

Choice data for domain 31, Coffee

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

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## Warning in tritrafo(x, y, z): negative components
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